The documentation and process conversion measures necessary to comply with this revision shall be completed by 9 Mar 93 .

INCH-POUND

MIL-S-19500/507B
9 December 1992
SUPERSEDING
MIL-S-19500/507A
14 September 1981

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, BIPOLAR TRANSIENT VOLTAGE SUPPRESSOR
TYPES 1N6036A THROUGH 1N6072A
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for 1500 watt peak pulse power transient voltage suppressor diodes. Four levels of product assurance are provided for each device as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings. Maximum ratings are as shown in columns 5 through 8 of table III herein, and as follows:

P_{PP} = 1500 W (see figure 3) at t_p = 1.0 ms.

$P_{M(AV)}$ = 1.0 W (derate at 6.67 mW/°C above T_A = +25°C) (see 6.3).

-55°C ≤ T_{OP} ≤ +175°C (ambient), -55°C ≤ T_{STG} ≤ +175°C (ambient).

1.4 Primary electrical characteristics at T_A = +25°C. Primary electrical characteristics are shown in columns 2 and 4 of table III herein.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500 and as follows:

V_C - - - - - - - - - - - - Clamping voltage. The maximum peak voltage appearing across the device when subjected to the peak pulse current I_P.

I_{PP} - - - - - - - - - - - - Peak pulse current at a specified condition.

$I_{(BR)}$ - - - - - - - - - - - - Reverse breakdown current at a specified condition.

P_{PP} - - - - - - - - - - - - Repetitive peak pulse power.

$\alpha V_{(BR)}$ - - - - - - - - - - - - Temperature coefficient of $V_{(BR)}$.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and figure 1 herein.

3.3.1 Metallurgical bond construction. Metallurgically bonded construction is required. The bonding metal shall have flow points above 260°C.

3.3.2 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750 and MIL-S-19500. Where choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Devices shall be marked as specified in MIL-S-19500. At the option of the manufacturer the following marking may be omitted from the body of the device:

a. Country of origin.

b. Polarity (this is a bipolar device).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein except, lot accumulation period shall be six months in lieu of six weeks.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JANS, JANTX, AND JANTXV levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3	$T_{(high)}$ = +175°C	$T_{(high)}$ = +175°C
9	Not applicable	Not applicable
10	Not applicable	Not applicable
11	Not applicable	Not applicable
12	See 4.5.1	See 4.5.1
13	Interim electrical, delta, and group A subgroup 2 electrical parameters not applicable for this screen (performed in screen 12). Subgroup 3 of table I herein.	Interim electrical, delta, and group A subgroup 2 electrical parameters not applicable for this screen (performed in screen 12).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. End-point electrical measurements shall be in accordance with the applicable steps of table II herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

a. Subgroup 3: Condition for surge is 1 cycle, (see 4.5.3b).

b. Subgroup 7: Condition for accelerated steady-state operation life are as follows: See 4.5.1, 1 ms pulse only (see 4.5.3.b), T_A = +100°C (min), T_J = +225°C (min), t = 168 hours (min).

c. Subgroups 4 and 6: Not applicable.

NOTE: All electrical measurements and test shall be performed twice, one in each direction.

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4.4.2.2 Group B inspection, table IVb (JAN, JANTX and JANTXV) of MIL-S-19500.

a. Subgroup 2: Condition for surge is 1 cycle, (see 4.5.3b).

b. Subgroup 3: Condition for steady-state operation life are as follows: See 4.5.2, 1 ms pulse only (see 4.5.3.b).

c. Subgroup 5: Not applicable.

NOTE: All electrical measurements and test shall be performed twice, one in each direction.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

a. Subgroup 2: Condition for terminal strength are as follows: Lead tension: Test condition A; weight = 5 pounds; t = 15 ±3 s. Lead fatigue: Test condition E; weight = 8 ounces.

b. Subgroup 6: Condition for steady-state operation life are as follows: See 4.5.1, 1 ms pulse only (see 4.5.3.b).

c. Subgroup 7: Condition for temperature coefficient of breakdown voltage are as follows: $I_{(BR)}$ = column 3 of table III, T_1 = +25°C ±3°C, T_2 = T_1 +100°C, LTPD = 10.

d. Subgroup 8: Condition for maximum peak pulse current are as follow: See 4.5.3a (20 μs pulse only) 10 pulses in each polarity, LTPD 10.

NOTE: All electrical measurements and test shall be performed twice, one in each direction.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows:

4.5.1 Power burn-in (HTRB) and steady-state operation life test conditions. The test conditions and order of events shall be as follows:

a. Read and record I_D and $V_{(BR)}$ in polarities A and B at T_A = +25°C.

b. Pulse in accordance with 4.5.3b, in polarity A 10 times (screening and group B) and 50 times (group C) at T_A = +25°C.

c. Pulse in accordance with 4.5.3b, in polarity B 10 times (screening and group B) and 50 times (group C) at T_A = +25°C.

d. Read I_D in polarities A and B at T_A = +25°C, remove defective devices and record the number of failures.

e. Apply the working peak reverse voltage (V_{WM}) (column 4 of table III) in polarity A at T_A = +125°C as follows:

1. 48 hours (JANTX and JANTXV) and 120 hours (JANS) for the screening test.

2. 170 hours (JAN, JANTX, and JANTXV) for group B steady-state operation life test.

3. 500 hours for group C steady-state operation life test.

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f. Read I_D in polarity A at T_A = +25°C. Devices with ΔI_D > 50 percent (100 percent for steady-state operation life) of the initial reading or 1 μA dc, whichever is greater, shall be considered defective. Remove defective devices and record the number of failures (see note).

g. Apply the working peak reverse voltage (V_{WM}) (column 4 of table III) in polarity B at T_A = +125°C as follows:

1. 48 hours (JANTX and JANTXV) and 120 hours (JANS) for the screening test.

2. 170 hours (JAN, JANTX, and JANTXV) for group B steady-state operation life test.

3. 500 hours for group C steady-state operation life test.

h. Read I_D = in polarity B at T_A = +25°C. Devices with ΔI_D > 50 percent (100 percent for steady-state operation life) of the initial reading or 1 μA dc, whichever is greater, shall be considered defective. Remove defective devices and record the number of failures (see note).

i. Read $V_{(BR)}$ in polarities A and B at T_A = +25°C. Devices with $\Delta V_{(BR)}$ > ±2 percent (±5 percent for steady-state operation life) of the initial reading shall be considered defective. Remove defective devices and record the number of failures.

j. Read I_D in polarity A at T_A = +25°C, remove defective devices and record the number of failures (see note).

NOTE: For the purposes of this test, the direction in which the device is first pulsed shall be considered polarity A and the reverse direction polarity B. For steady-state operation life, I_D maximum limits shall be two times the group A limit.

4.5.1.1 Group C steady-state operation life test (alternate procedure). When the group B 340 hour life test is continued on test to 1,000 hours to satisfy the group C life test requirements, the test shall be performed as given in 4.5.1 with the following exceptions:

a. 4.5.1, steps b and c shall be moved and performed following step h.

b. 4.5.1, steps f and h shall be repeated after steps b and c are performed and before step i is completed (step j may be omitted when this procedure is used).

4.5.2 Accelerated steady-state operation life. This test shall be conducted with the devices subjected to the breakdown current specified in column 10 of table III in opposite polarities for 84 +8, -4 hours in each polarity. At the beginning of the test and at the end of each time period, the devices shall be temperature stabilized at T_A = 25°C and subjected to pulse conditions at the rate of one pulse per minute (max) for 10 pulses each in accordance with 4.5.3 as specified.

4.5.3 Maximum peak pulse current (I_{pp}). The peak pulse currents specified in column 7 of table III shall be applied while simultaneously maintaining a bias voltage, not less than the applicable voltage in column 4 of table III, in the same polarity as the peak pulse current. The peak pulse current shall be applied with a current vs time waveform as follows (1 pulse per minute maximum):

a. Pulse current shall reach 100 percent of I_{pp} at t ≤ 8 μs and decay to 50 percent of I_{pp} at t ≥ 20 μs (see figure 5).

b. Pulse current shall reach 100 percent of I_{pp} at t ≤ 10 μs and decay to 50 percent of I_{pp} at t ≥ 1 ms for t_p = 1 ms (see figure 4).

4.5.4 Clamping voltage. The peak pulse clamping voltage shall be measured across the diode in a 1 ms time interval. The response detector shall demonstrate equipment accuracy of ±3 percent. The peak clamping voltage as specified in column 6 of table III shall be applicable to the 1 ms pulse of 4.5.3b only.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits 2/		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2 3/						
Reverse current	4016	DC method; $V_R = V_{WM}$ (column 4 of table III)	I_R		column 5	µA dc
Breakdown voltage	4022	$t_p \leq 300$ ms, duty cycle ≤ 2%; $I_{(BR)}$ = column 3 to table III	$V_{(BR)}$	column 2	column 2	V dc
Subgroup 3 3/						
Minimum breakdown voltage	4022	$t_p \leq 300$ ms, duty cycle ≤ 2%; $I_{(BR)}$ = column 3 of table III $T_A = -55°C$	$V_{(BR)}$	column 9		V dc
Subgroup 4 3/						
Clamping voltage maximum (pulsed) (see 4.5.4)		$t_p = 1.0$ ms (see 4.5.3.b); I_{PP} = column 7 of table III	V_C		column 6	V (pk)
Subgroup 5 and 6						
Not applicable						

1/ For sampling plan, see MIL-S-19500.

2/ Column references are to table III.

3/ All electrical testing shall be performed twice, once in each direction.

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TABLE II. Groups B and C electrical measurements. 1/ 2/ 3/ 4/

Step	Inspection	MIL-STD-750		Symbol	Limits 5/		Unit
		Method	Conditions		Min	Max	
1	Reverse current	4016	DC method, $V_R = V_{WM}$ column 4 of table III	I_R		column 5	µA dc
2	Breakdown voltage	4022	$t_p \leq 300$ ms; duty cycle ≤ 2%; $I_{(BR)}$ = column 3 of table III	$V_{(BR)}$	column 2	column 2	V dc
3	Standby current	4016	DC method, $V_R = V_{WM}$ (column 4 of table III	ΔI_D 6/		100% of initial reading or 20% of column 5 of table III, whichever is greater.	
4	Breakdown voltage	4022	$t_p \leq 300$ ms; duty cycle ≤ 2%; $I_{(BR)}$ = column 3 of table III	$\Delta V_{(BR)}$ 6/		±5% of initial value	
5	Clamping voltage		$t_p = 1.0$ ms (see 4.5.3.b) I_p column 7 of table III	V_C		column 6	V (pk)

1/ All electrical testing shall be performed twice, once in each direction.

2/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroup 3, see table II herein, steps 1, 2, and 5.
b. Subgroup 7, see table II herein, steps 1, 2, 3, and 4.

3/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 2.
b. Subgroup 6 see table II herein, steps 1, 2, 3, and 4.

4/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroup 2 and 3, see table II herein, steps 1, and 2 for JANS and steps 1 and 2 for JANTX and JANTXV.
b. Subgroup 8, see table II herein, steps 1 and 2 for all levels.

5/ Column references are to table III.

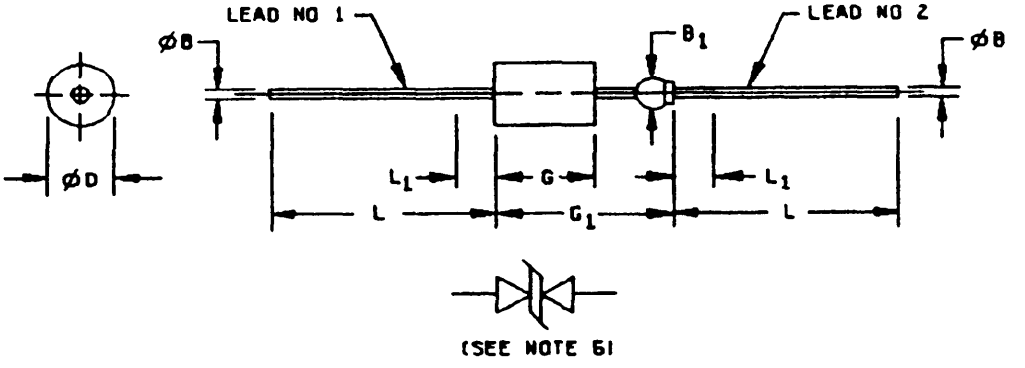
6/ Devices which exceed the group A limits for this test shall not be accepted.

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TABLE III. Characteristics and Ratings.

Col 1	Col 2		Col 3	Col 4	Col 5	Col 6	Col 7		Col 8	Col 9	Col 10
Type	Breakdown voltage $V_{(BR)}$ at I_{IBR}		Test current $I_{(BR)}$	Working peak reverse voltage V_{WM}	Maximum reverse current I_R	Maximum clamping voltage V_C at I_{PP}	Maximum peak pulse current (I_{PP})		Maximum temperature coefficient of $V_{(BR)}$ $\alpha V_{(BR)}$	Minimum breakdown voltage at $I_{(BR)}$ T_A = -55°C	Breakdown current $I_{(BR)}$ maximum dc current T_A = +25°C
	Min	Max					t_p = 20 µs t_r = 8 µs	t_p = 1 ms t_r = 10 µs			
	V dc	V dc	mA dc	V (pk)	µA dc	V (pk)	A (pk)	A (pk)	α/°C	V dc	mA dc
1N6036A	7.13	7.88	10	6.0	1000	11.3	750	132.0	.061	6.66	125.0
1N6037A	7.79	8.61	10	7.0	500	12.1	700	124.0	.065	7.24	115.0
1N6038A	8.65	9.55	10	7.5	200	13.4	630	112.0	.068	8.01	104.0
1N6039A	9.50	10.50	1	8.5	50	14.5	585	103.0	.073	8.75	95.0
1N6040A	10.50	11.60	1	9.0	10	15.6	545	96.0	.075	9.65	86.0
1N6041A	11.40	12.60	1	10.0	5	16.7	510	90.0	.078	10.40	73.0
1N6042A	12.40	13.70	1	11.0	5	18.2	465	82.0	.081	11.30	70.0
1N6043A	14.30	15.80	1	12.0	5	21.2	400	71.0	.084	13.00	63.0
1N6044A	15.20	16.80	1	13.0	5	22.5	375	67.0	.086	13.70	59.0
1N6045A	17.10	18.90	1	15.0	5	25.2	335	59.5	.088	15.40	53.0
1N6046A	19.00	21.00	1	17.0	5	27.7	305	54.0	.090	17.10	47
1N6047A	20.90	23.10	1	18.0	5	30.6	275	49.0	.092	18.80	4
1N6048A	22.80	25.20	1	20.0	5	33.2	255	45.0	.094	20.50	39
1N6049A	25.70	28.40	1	22.0	5	37.5	225	40.0	.096	23.00	35.0
1N6050A	28.50	31.50	1	25.0	5	41.4	205	36.0	.097	25.50	31.0
1N6051A	31.40	34.70	1	28.0	5	45.7	185	33.0	.098	29.00	28.0
1N6052A	34.20	37.80	1	30.0	5	49.9	170	30.0	.099	30.50	26.0
1N6053A	37.10	41.00	1	33.0	5	53.9	155	28.0	.100	33.10	24.0
1N6054A	40.90	45.20	1	36.0	5	59.3	145	25.3	.101	36.40	22.0
1N6055A	44.70	49.40	1	40.0	5	64.8	130	23.2	.101	39.80	20.0
1N6056A	48.50	53.60	1	43.0	5	70.1	120	21.4	.102	43.10	18.0
1N6057A	53.20	58.80	1	47.0	5	77.0	110	19.5	.103	47.30	17.0
1N6058A	58.90	68.20	1	53.0	5	85.0	100	17.7	.104	52.30	15.0
1N6059A	64.60	71.40	1	58.0	5	92.0	90	16.3	.104	57.30	14.0
1N6060A	71.30	78.80	1	64.0	5	103.0	82	14.6	.105	63.20	12.0
1N6061A	77.90	86.10	1	70.0	5	113.0	75	13.3	.105	69.00	11.0
1N6062A	86.50	95.50	1	75.0	5	125.0	68	12.0	.106	76.50	10.0
1N6063A	95.00	105.00	1	82.0	5	137.0	62	11.0	.106	84.10	9.5
1N6064A	105.00	116.00	1	94.0	5	152.0	55	9.9	.107	92.80	8.5
1N6065A	114.00	126.00	1	100.0	5	168.0	50	8.9	.107	100.00	7.5
1N6066A	124.00	137.00	1	110.0	5	182.0	47	8.2	.107	109.00	7.0
1N6067A	143.00	158.00	1	128.0	5	213.0	40	7.0	.108	126.00	6.0
1N6068A	162.00	179.00	1	145.0	5	245.0	36	6.1	.108	143.00	5.5
1N6069A	171.00	189.00	1	150.0	5	261.0	34	5.7	.108	151.00	5.2
1N6070A	181.00	200.00	1	160.0	5	278.0	32	5.4	.108	157.00	5.0
1N6071A	190.00	210.00	1	170.0	5	294.0	30	5.1	.108	169.00	4.7
1N6072A	209.00	231.00	1	185.0	5	328.0	25	4.6	.108	184.00	4.3

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
ØB	.025	.035	0.64	0.89	
B_1	.045	.100	1.14	2.54	3
ØD	.215	.235	5.46	5.97	1
G	.293	.357	7.44	9.07	
G_1	----	.570	----	14.48	
L	1.000	1.625	25.40	41.28	
L_1	----	.188	----	4.78	2

NOTES:
1. The major diameter is essentially constant along its length.
2. Within this zone, diameter may vary to allow for lead finishes and irregularities.
3. Dimension to allow for pinch or seal deformation anywhere along tubulation.
4. Lead 1 is electrically connected to the case.
5. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
6. Symbol for internal construction of a bipolar transient suppressor.

FIGURE 1. Physical dimensions, type 1N6036A through 1N6072A (DO-13).

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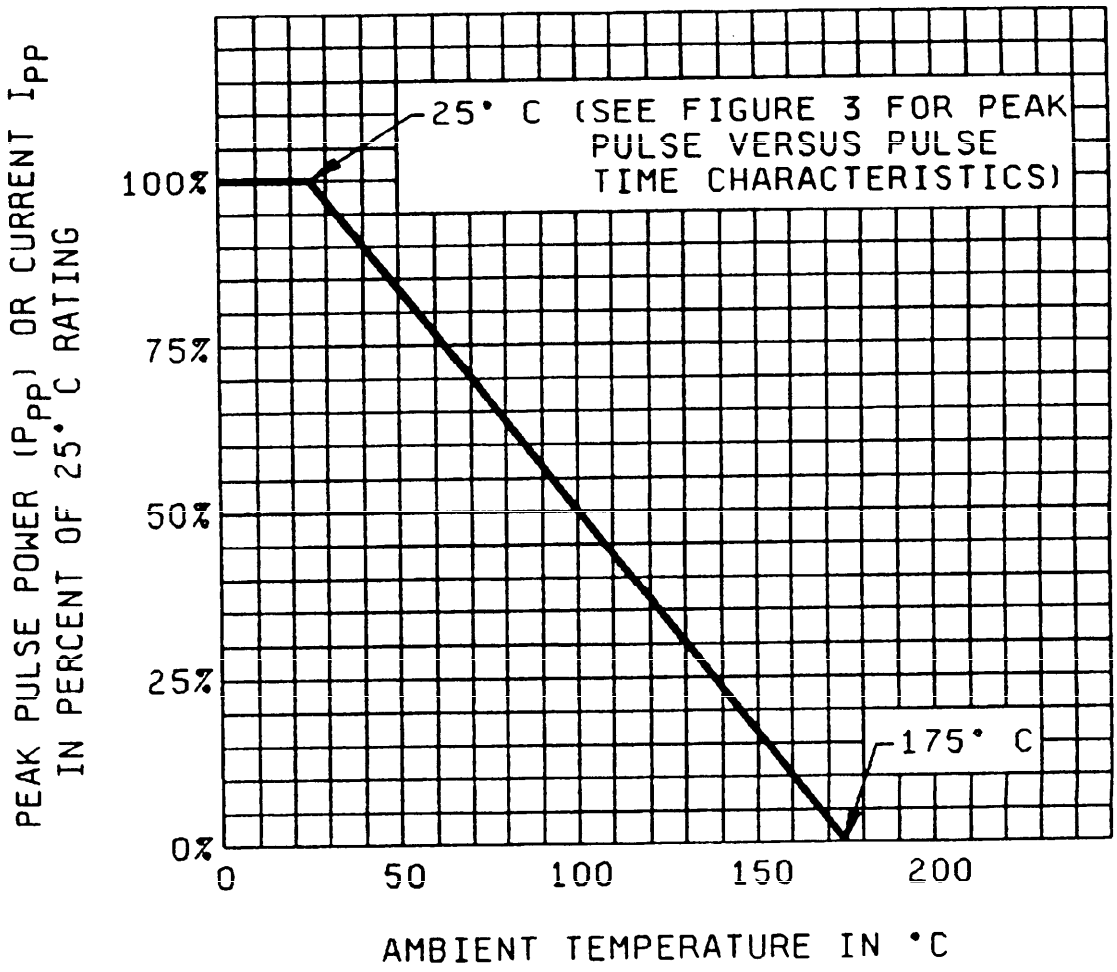


FIGURE 2. Derating curve.

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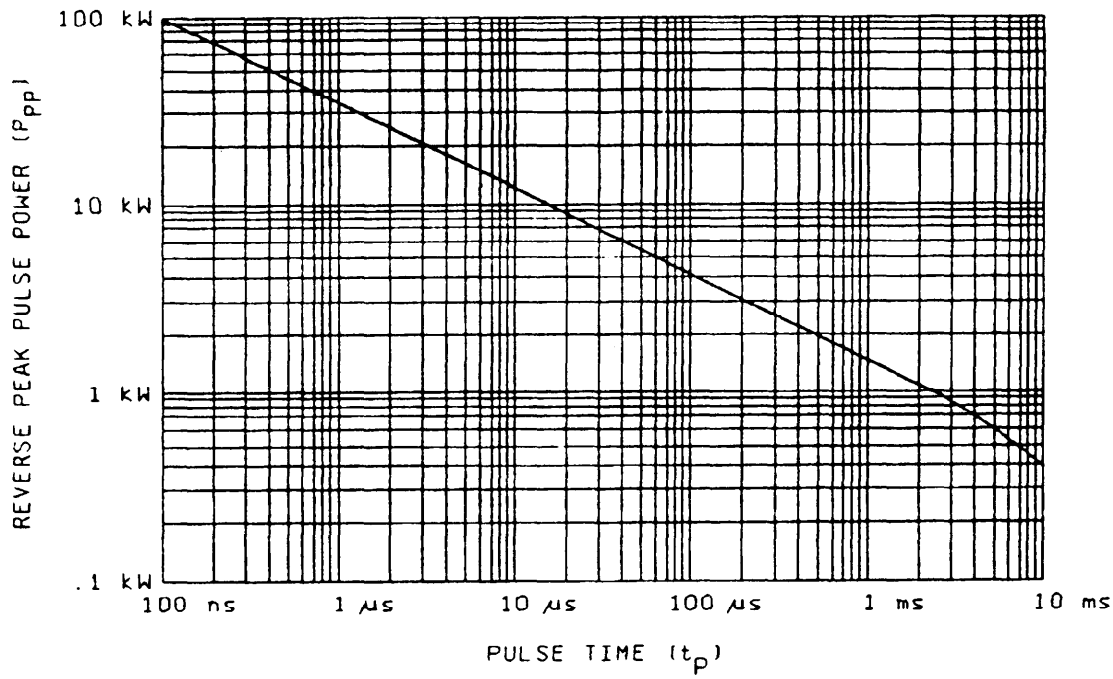
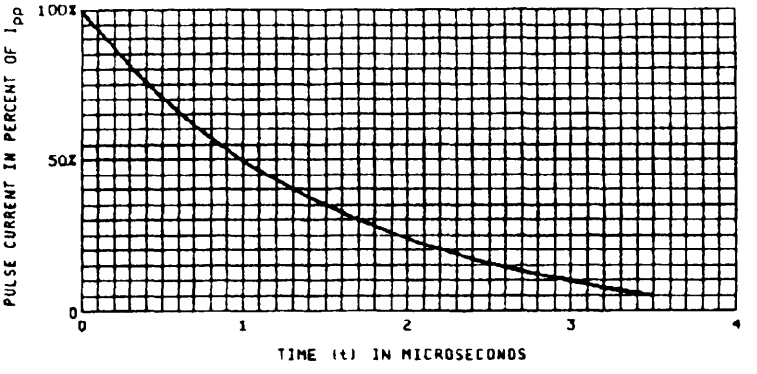


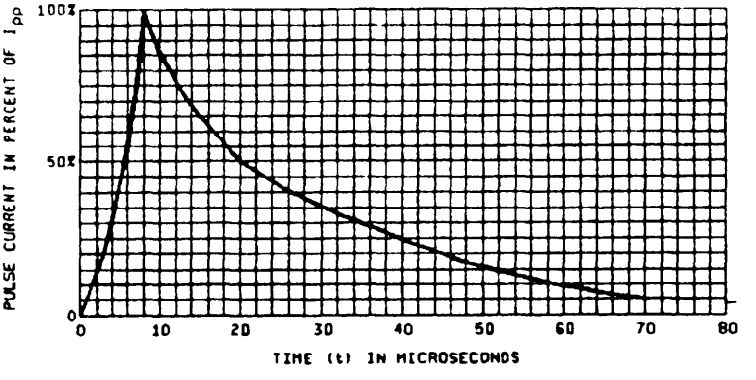
FIGURE 3. Peak pulse power vs. pulse time.

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NOTE: Pulse time duration is defined as that point where the pulse current decays to 50 percent of I_{pp}. (Rise time to 100 percent of I_{pp} = 10 μs)

FIGURE 4. Current impulse waveform.



NOTE: Pulse time duration is defined as that point where the pulse current decays to 50 percent of I_{pp}. (Rise time to 100 percent of I_{pp} = 8 μs)

FIGURE 5. Current impulse waveform.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.2).

c. Type designation and product assurance level.

6.3 Steady-state power rating. This rating is not relevant for most-applications.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - MI
Air Force - 19, 99
DLA - ES

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5961-1389)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:34Z
Check the source to verify that this is the current version before use.

measured shall not have changed during the 101 Naid-bias test from those recorded in 4.5.10 by